

LALITH SAGAR KAMBALA

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SUMMARY

Full Stack & AI Software Engineer who builds with ownership from architecture decisions to production deployments. I gravitate toward hard problems at the intersection of AI and product, turning ambiguous requirements into reliable, scalable systems that real users depend on. Driven by outcomes over output, I care about why something is built as much as how.

EDUCATION

Master of Science: Computer Science

Aug 2023 to May 2025

University of Central Missouri - Lee's Summit, MO – **GPA:** 3.55/4.00

EXPERIENCE

Founding Software Engineer

Sep 2025 to Present

Todaiyo.ai – MA, United States

- Architected and built a production-grade full-stack platform from 0→1 using React, TypeScript, Node.js and Python, designing microservice architecture with Supabase (PostgreSQL + Edge Functions) for scalable data handling, implementing REST API integrations across HubSpot, Attom, and third-party services, and owning end-to-end deployment via CI/CD pipelines on Netlify servers.
- Engineered a multi-modal AI pipeline integrating OpenAI LLMs for text, voice, and image processing with natural language understanding, orchestrated vector embeddings and RAG pipelines to extract and analyze large-scale data from files and external APIs and built an agentic backend with intent-based routing to dynamically adapt user flows in production.
- Designed and implemented scalable payment and subscription infrastructure using Stripe gateway and In-App Purchase (IAP) flows, developed responsive, user-facing web pages and app screens with optimized UX flows, and integrated microservice-level business logic to support multi-platform product delivery.
- Implemented Google Analytics 4 (GA4) and Google Tag Manager (GTM) with 75%+ event coverage across the platform, instrumented custom tracking and conversion funnels, and leveraged usage analytics to drive product engagement improvements and inform feature prioritization decisions.

AI Full Stack Engineer (*Part-time*)

Sep 2025 to Present

Consumer Genie – United States

- Architected a full-stack B2B multi-agent intelligence platform using React, TypeScript, and Tailwind CSS on the frontend with Python, Django, and Node.js on the backend, designed Figma-based UI/UX flows for merchant-facing dashboards, implemented GraphQL and REST APIs, and deployed scalable microservices on AWS (EC2, RDS, S3) with CloudWatch monitoring and CI/CD pipelines for automated backend delivery.
- Built a production-grade agentic AI system using LangChain, LangGraph, and MCP (Model Context Protocol) for multi-agent orchestration, fine-tuned LLMs for review analysis and product ranking agents, engineered inference optimization with accuracy-latency tradeoff evaluation, and integrated OpenAI and Claude APIs with Pydantic-validated structured outputs, robust error handling, and model fallback strategies for context-aware decision pipelines.
- Engineered semantic search and vector search pipelines using MongoDB with embedding models, implemented Kafka-based event-driven data ingestion for concurrent crawling, validation, and summarization workloads at scale, and optimized PostgreSQL query performance through schema design, indexing, and Redis caching strategies to support high-throughput consumer product data processing.
- Delivered Dockerized microservices deployments on AWS with CI/CD pipelines, monitored production systems via AWS CloudWatch with structured logging, distributed tracing, and alerting on agent inference latency, token throughput, and error rates as key performance metrics, and maintained code quality through Git-based reviews, integration test coverage, and technical documentation.

Full Stack Software Engineer

Jul 2025 to Oct 2025

IEJL – CA, United States

- Developed and maintained responsive web applications using Next.js, React, TypeScript, and Tailwind CSS, translating Figma designs into pixel-perfect, accessible interfaces with optimized UX flows, and integrated RESTful and micro-APIs across a microservice architecture to support scalable production workflows.
- Implemented end-to-end authentication and authorization infrastructure using OAuth2, JWT, and secure session management, enforced API security across microservices with role-based access control (RBAC) and privacy compliance standards, and managed Supabase as the backend data layer with optimized schema design.
- Designed and optimized data workflows and system integrations to support analytics, reporting, and operational decision-making, enabling efficient data retrieval for downstream AI-driven and business intelligence systems, and partnered with product managers and UI/UX designers in Agile sprints to reduce iteration cycles by 35%
- Deployed and managed containerized services using Docker on AWS, implemented CI/CD pipelines to streamline release workflows, and improved system reliability by 30% through structured logging, environment configuration management, and cloud infrastructure best practice.

Web Development Intern

Jun 2022 to Dec 2022

Coincent Technologies – India

- Developed and maintained backend integrations between EHR systems and healthcare appointment platforms, improving appointment processing efficiency by 35% through optimized REST API workflows and data synchronization.

- Engineered scalable backend services and database query optimizations that reduced data inconsistencies by 40% while supporting real-time patient record updates and appointment scheduling across multiple healthcare systems.
- Implemented secure API integration and backend consistency checks for healthcare applications, improving system reliability by 30% and enabling seamless management of 10,000+ appointment transactions and patient record requests.

CORE COMPETENCIES

- Languages & Frameworks: Python, React/Next.js, JavaScript/TypeScript, Node.js, HTML/CSS, Tailwind CSS, Flask, Express, SpringBoot, Pydantic
- Databases: MongoDB, PostgreSQL, Supabase, Redis, MySQL, Vector DB – Pinecone, FAISS
- Real-Time & Event-Driven: Real-Time & Event-Driven: System Design, WebSockets, Kafka, event-driven microservices, async processing, distributed systems, relational databases, RAG Pipelines, Chat Orchestration, SEO, custom analytics/tracking
- API Integration: RESTful APIs, ERP/CRM systems, GraphQL, Redis, Stripe, In-App purchases(IAP), HubSpot, OpenAI, Claude API, Web Scraping (DDG/Serper), Tavily
- Design & UX: Figma (UI/UX design, wireframing, app screen design), responsive design, design systems, component libraries, UX design principles, Core Web Vitals
- Security: OAuth2, JWT, authentication, authorization, privacy, secure deployment
- AI/ LLM: LangChain, LangGraph, Embeddings, Fine-tuning, MCP, ML Inference, RAG pipelines, LLM-agnostic architecture, Agentic orchestration, Prompt Engineering
- Cloud & DevOps: AWS (EC2, RDS, S3/CloudFront), Docker, Kubernetes, CI/CD, Git (code reviews), GCP, Jira, Agile/Scrum, Linux, Netlify, Vercel
- Testing & Observability: Unit/Integration testing, CloudWatch, Logging, Error Monitoring, GA4 Analytics, GTM

ACHIEVEMENTS & CERTIFICATIONS

- Anthropic - MCP, Agents & Sub-Agents (Advanced)
- Oracle Cloud Infrastructure Generative AI Professional (1Z0-1127-24)
- Introduction to Large Language Models (LLM) by Google Cloud
- Accenture North America Data Science and Visualization Job Simulation on Forage

PROJECTS

AI- Debugging Agent using ollama – Pinecone

Mar 2026

- Built a modular agentic debugging pipeline with CodeLlama fine-tuned via RAG (Pinecone), featuring retrieval orchestration, structured outputs, namespace-based multi-repo querying, and privacy-first local vector store mode for on-device repository data.
- Designed evaluation frameworks with multi-agent RFT techniques, measuring reasoning quality, inference latency, throughput, and reliability; calculated entropy differentials across model variants and implemented production-grade logging, LLM/Pinecone fallback strategies, and query tracking across production workloads.

Genie Shop

Sep 2025

- Developed a product search system using LangGraph state machine with intent-based routing that processes research and specific product queries through separate pipelines, streaming real-time results via Streamlit.
- Built web scraping pipeline using Google Gemini for intent classification and product extraction, integrating DuckDuckGo/Serper API to search across major retailers with URL scoring and filtering mechanisms.
- Created location-aware cross-platform retrieval system with category-based site selection across USA markets, generating automated fallback search URLs and implementing multi-model retry strategies that ensure query completion rate despite API failures.

Article Semantic Search using LLM's

Aug 2025

- Designed a financial news research platform leveraging LangChain's Unstructured URL Loader and OpenAI embeddings for precise content indexing and retrieval.
- Deployed FAISS-based semantic search with embedding vector storage, and built an interactive Streamlit app allowing users to process URLs and obtain contextually accurate answers from an LLM-driven pipeline.

Brain Tumor Detection Using Deep Learning

Jan 2023 - Apr 2023

- Constructed a CNN model with Conv2D, Batch Normalization, MaxPooling, Dropout, and Dense layers using ReLU activation to classify MRI scans with 94% accuracy, enhancing feature extraction and reducing overfitting by 15%.
- Automated early tumor detection workflows, cutting diagnosis time by up to 40% and improving patient care through scalable, real-time image analysis.

Blockchain project on Property Management System

Aug 2022 - Dec 2022

- Built **Block Wizz**, a decentralized blockchain-based tax payment solution that processed 500+ transactions with 100% transparency, ensuring tamper-proof records and full auditability across the network.
- Incorporated SHA-256 hashing to securely chain payment histories, reducing fraudulent receipts and duplicate payments by 95% while enabling real-time, verifiable transaction tracking.